Enterprise AI Architect Learning Path v3 – Phase Guide 6

Enterprise AI Architect Learning Path v3 – Phase Guide 6

AI Publishing Company Platform

Purpose of This Phase

The previous phases established:

- Engineering foundations
- Production RAG architectures
- Agent orchestration patterns
- MCP-based tool ecosystems

Phase 6 introduces the final major application architecture of the learning path:

> Designing and operating a collaborative multi-agent system.

The objective is not to build a writing assistant.

The objective is to build a structured, governed, long-running workflow that demonstrates how multiple specialized AI agents can cooperate while remaining under human control.

The AI Publishing Company serves as a practical implementation of enterprise workflow orchestration, where distinct agents perform specialized functions and collaborate to produce a business outcome.

Why This Phase Matters

Most AI systems today are:

- Single-agent
- Prompt-driven
- Session-based

Enterprise systems increasingly require:

- Multiple agents
- Persistent workflows
- Human oversight
- Version management
- Long-lived state
- Role separation

This phase demonstrates how to architect systems where agents become organizational participants rather than isolated utilities.

Phase Objectives

Upon completion of this phase, you should be able to:

- Design multi-agent workflows
- Define agent responsibilities
- Implement role boundaries
- Coordinate long-running processes
- Manage workflow state
- Maintain version history
- Implement approval processes
- Separate creative and review functions
- Build enterprise workflow architectures
- Design human-governed agent ecosystems

Architect Mindset

One of the most important lessons of this phase is:

> Not every problem should be solved by one agent.

A common mistake is creating a single "super agent" responsible for everything.

Enterprise architecture generally favors:

Specialized Components

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Clear Responsibilities

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Controlled Interactions

The same principle applies to agent systems.

The question is not:

> How capable can one agent become?

The question is:

> How should responsibilities be distributed across a system?

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Primary Resources

Required

R1

AI Engineering

Focus Areas:

- Workflow design
- Evaluation
- Agent collaboration

R18

Generative AI with LangChain

Focus Areas:

- Multi-agent systems
- Workflow orchestration
- Agent coordination

R10

LangGraph Documentation

Focus Areas:

- StateGraph design
- Persistence
- Long-running workflows

R11

LangGraph with Azure AI Foundry

Focus Areas:

- Enterprise deployment patterns
- Agent services

R8

Semantic Kernel Documentation

Focus Areas:

- Agent integration
- Tool usage
- Workflow participation

Core Technical Skills

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1. Multi-Agent Architecture

Learn to define:

Agent Roles

Each agent should have:

- A clear responsibility
- Limited authority
- Well-defined outputs

Avoid:

- Overlapping agents
- Vague responsibilities
- Autonomous chaos

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2. Workflow Orchestration

Design systems using:

- States
- Transitions
- Checkpoints
- Approval gates

Workflows should be predictable and explainable.

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3. Persistent State Management

Workflows may span:

- Hours
- Days
- Weeks

State must survive:

- Restarts
- Interruptions
- Human feedback cycles

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4. Human Governance

Humans remain responsible for:

- Approval
- Direction
- Final decisions

Agents should support decision-making rather than replace accountability.

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5. Version Control & Traceability

Every major workflow decision should be traceable.

Examples:

- Draft versions
- Review comments
- Approval decisions
- Revision history

Enterprise systems often require the ability to reconstruct how an outcome was produced.

Designing the AI Publishing Company

The publishing platform serves as both:

A Product

A workflow that assists authors.

A Reference Architecture

A model for enterprise multi-agent systems.

Organizational Structure

Think of the system as a virtual organization.

Acquisition Agent

Responsibilities:

- Evaluate concepts
- Research opportunities
- Identify gaps

Outputs:

- Project proposals
- Opportunity assessments

Developmental Editor Agent

Responsibilities:

- Evaluate structure
- Evaluate organization
- Improve content planning

Outputs:

- Outline recommendations
- Structural improvement plans

Continuity Agent

Responsibilities:

- Verify consistency
- Identify contradictions
- Preserve long-term coherence

Outputs:

- Continuity reports
- Consistency findings

Copy Editor Agent

Responsibilities:

- Improve language quality
- Improve readability
- Improve clarity

Outputs:

- Revision recommendations
- Copy-edit reports

Publishing Coordinator Agent

Responsibilities:

- Coordinate workflow
- Manage state
- Track approvals
- Maintain records

Outputs:

- Workflow status
- Release readiness reports

Workflow Architecture

A typical workflow might resemble:

Idea

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Acquisition Review

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Outline Creation

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Development Review

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Draft Creation

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Continuity Review

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Copy Editing

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Human Approval

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Publish

This architecture should be implemented as a stateful workflow rather than a sequence of independent prompts.

Phase Project

AI Publishing Company v1

The goal is to build a functioning multi-agent workflow platform.

Functional Requirements

Idea-to-Outline Workflow

Support:

- Concept generation
- Evaluation
- Approval

Drafting Workflow

Support:

- Structured content creation
- Progress tracking
- State management

Review Workflow

Support:

- Developmental review
- Continuity review
- Copy editing

Approval Workflow

Support:

- Human review
- Revision requests
- Final approval

Workflow Tracking

Support:

- Current state
- History
- Revision tracking

Architecture Deliverables

ADR-016

Multi-Agent Strategy

Document:

- Why multiple agents were selected
- Agent responsibilities
- Workflow boundaries

ADR-017

Workflow Design

Document:

- State model
- Transition logic
- Approval processes

ADR-018

Human Governance Model

Document:

- Human responsibilities
- Approval points
- Escalation rules

Portfolio Artifacts

StateGraph Diagram

Illustrate:

Agents

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Workflow States

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Transitions

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Human Approval

Agent Responsibility Matrix

Define:

- Agent
- Responsibilities
- Inputs
- Outputs

Workflow Specification

Document:

- States
- Transitions
- Approval checkpoints

Governance Notes

Describe:

- Human control
- Review procedures
- Accountability model

Suggested Sprint Progression

Sprint 1 — Workflow Design

Goals:

- Define workflow
- Define agent roles

Done When:

- Architecture documented

Sprint 2 — State Model

Goals:

- Create workflow state machine
- Define persistence strategy

Done When:

- State architecture complete

Sprint 3 — Agent Development

Goals:

- Build individual agents
- Define interfaces

Done When:

- Core agents operational

Sprint 4 — Workflow Integration

Goals:

- Connect agents

- Enable transitions

Done When:

- Workflow executes end-to-end

Sprint 5 — Review Systems

Goals:

- Implement editorial workflows
- Implement continuity checks

Done When:

- Review process operational

Sprint 6 — Approval Workflows

Goals:

- Add governance controls
- Add human checkpoints

Done When:

- HITL controls functioning

Sprint 7 — Hardening

Goals:

- Improve reliability
- Improve traceability
- Improve documentation

Done When:

- AI Publishing Company v1 completed

Common Pitfalls

Pitfall 1

Creating Too Many Agents

More agents do not necessarily create a better architecture.

Specialization should solve a problem.

Pitfall 2

No Workflow Ownership

Some component should always know:

- Current state
- Next step
- Responsible agent

Pitfall 3

No Human Control

Enterprise systems require governance.

Human approval should be intentional rather than optional.

Pitfall 4

Allowing Agents to Rewrite Everything

Agent responsibilities should remain focused.

Excessive overlap creates unpredictable behavior.

Pitfall 5

Ignoring State Persistence

Long-running workflows require durable state management.

Without persistence, the system becomes unreliable.

Enterprise Translation Layer

The Publishing Company architecture maps directly to enterprise business workflows.

	Publishing Workflow	Enterprise Equivalent
1	Book Production	Product Development
2	Editorial Review	Compliance Review
3	Continuity Review	Quality Assurance
4	Publishing Coordinator	Workflow Orchestrator
5	Author Approval	Business Approval
6	Manuscript Lifecycle	Business Process Lifecycle

The underlying architectural patterns remain largely identical.

Definition of Done

Phase 6 is complete when:

- Multi-agent architecture is implemented
- Workflow state persists correctly
- Agent responsibilities are clearly defined
- Review workflows function correctly
- Human approval gates exist
- Governance model is documented
- Workflow diagrams are complete
- ADRs are completed
- AI Publishing Company v1 operates end-to-end

End-of-Phase Self-Assessment

You are ready for Phase 7 when you can confidently answer:

- Can I design a multi-agent architecture?
- Can I justify agent boundaries and responsibilities?
- Can I orchestrate long-running workflows?
- Can I persist workflow state?
- Can I implement human governance controls?
- Can I explain how multi-agent systems map to enterprise business processes?

If all answers are "yes," proceed to **Phase 7 – LLMOps, Observability & Deployment**.